# Latest Syllabus 2025-26

## (Issued by CBSE Dated on 28th March, 2025)

## MATHEMATICS STANDARD-(Code No. 041)

## Class-X-(Theory)

| Unit No. | Unit Name | Marks |
|---|---|---|
| I | Number Systems | 06 |
| II | Algebra | 20 |
| III | Coordinate Geometry | 06 |
| IV | Geometry | 15 |
| V | Trigonometry | 12 |
| VI | Mensuration | 10 |
| VII | Statistics & Probability | 11 |
| | **Total** | **80** |

| S. No. | Content | Competencies | Explanation |
|---|---|---|---|
| | **UNIT I : NUMBER SYSTEMS** | | |
| 1. | **REAL NUMBERS**<br>1. Fundamental Theorem of Arithmetic - statements after reviewing work done earlier and after illustrating and motivating through examples.<br>2. Proofs of irrationality of $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. | • Develops understanding of numbers, including the set of real numbers and its properties.<br>• Extends the understanding of powers (radical powers) and exponents.<br>• Applies Fundamental Theorem of Arithmetic to solve problems related to real life contexts. | • Describes Fundamental Theorem of Arithmetic with examples<br>• Prove algebraically the Irrationality of numbers like $\sqrt{2}, \sqrt{3}, \sqrt{5}, 3+2\sqrt{5}$ etc. |
| | **UNIT II : ALGEBRA** | | |
| 1. | **POLYNOMIALS** | | |
| | 1. Zeros of a polynomial.<br>2. Relationship between zeros and coefficients of quadratic polynomials. | • Develops a relationship between algebraic and graphical methods of finding the zeroes of a polynomial. | • Find the zeros of polynomial graphically and algebraically and verifying the relation between zeros and coefficients of quadratic polynomials. |

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| **2.** | **PAIR OF LINEAR EQUATIONS IN TWO VARIABLES**<br>1. Pair of linear equations in two variables and graphical method of their solution, consistency/ inconsistency.<br>2. Algebraic conditions for number of solutions.<br>3. Solution of a pair of linear equations in two variables algebraically - by substitution, by elimination. Simple situational problems. | • Describes plotting a pair of linear equations and graphically finding the solution.<br>• Models and solves contextualised problems using equations (e.g., simultaneous linear equations in two variables). | • Find the solution of pair of linear equations in two variables graphically and algebraically (substitution and elimination method) |
| **3.** | **QUADRATIC EQUATIONS**<br>1. Standard form of a quadratic equation $ax^2 + bx + c = 0$, $(a \neq 0)$.<br>2. Solutions of quadratic equations (only real roots) by factorisation, and by using quadratic formula. Relationship between discriminant and nature of roots.<br>3. Situational problems based on quadratic equations related to day to day activities to be incorporated. | • Demonstrates strategies of finding roots and determining the nature of roots of a quadratic equation. | • Solves quadratic equations using factorisation and quadratic formula<br>• Determines the nature of roots using discriminant<br>• Formulates and solves problems based on real life context |
| **4.** | **ARITHMETIC PROGRESSIONS**<br>1. Motivation for studying Arithmetic Progression.<br>2. Derivation of the nth term and sum of the first n terms of A.P. and their application in solving daily life problems. | • Develops strategies to apply the concept of A.P. to daily life situations. | • Applies concepts of AP to find the nth term and sum of n terms.<br>• Application of AP in real life problems |
| | **UNIT III : COORDINATE GEOMETRY** | | |
| **1.** | **COORDINATE GEOMETRY**<br>**1. Review:** Concepts of coordinate geometry. Distance formula. Section formula (internal division). | • Derives formulae to establish relations for geometrical shapes in the context of a coordinate plane, such as, finding the distance between two given points, to determine the coordinates of a point between any two given points. | • Solves problems using distance formula and section formula |

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| UNIT IV: GEOMETRY | | | |
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| 1. | **TRIANGLES**<br>Definitions, examples, counter examples of similar triangles.<br>1. **(Prove)** If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.<br>2. **State (without proof)** If a line divides two sides of a triangle in the same ratio, the line is parallel to the third side.<br>3. **State (without proof)** If in two triangles, the corresponding angles are equal, their corresponding sides are proportional and the triangles are similar.<br>4. **State (without proof)** If the corresponding sides of two triangles are proportional, their corresponding angles are equal and the two triangles are similar.<br>5. **State (without proof)** If one angle of a triangle is equal to one angle of another triangle and the sides including these angles are proportional, the two triangles are similar. | • works out ways to differentiate between congruent and similar figures.<br>• establishes properties for similarity of two triangles logically using different geometric criteria established earlier such as, Basic Proportionality Theorem, etc. | • Prove Basic Proportionality theorem and applying the theorem and its converse in solving questions<br>• Prove similarity of triangles using different similarity criteria |
| 2. | **CIRCLES**<br>Tangent to a circle at, point of contact<br>1. **(Prove)** The tangent at any point of a circle is perpendicular to the radius through the point of contact.<br>2. **(Prove)** The lengths of tangents drawn from an external point to a circle are equal. | • derives proofs of theorems related to the tangents of circles. | • Prove the theorems based on the tangent to a circle.<br>• Applies the concept of tangents of circle to solve various problems. |

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| | **UNIT V: TRIGONOMETRY** | | |
| **1.** | **INTRODUCTION TO TRIGONOMETRY**<br>1. Trigonometric ratios of an acute angle of a right-angled triangle. Proof of their existence (well defined)<br>2. Motivate the ratios whichever are defined at 0° and 90° . Values of the trigonometric ratios of 30° , 45° and 60°.<br>3. Relationships between the ratios. | • Understands the definitions of the basic trigonometric functions (including the introduction of the sine and cosine functions). | • Evaluates trigonometric ratios<br>• Describes trigonometric ratios of standard angles and solving related expressions |
| **2.** | **TRIGONOMETRIC IDENTITIES**<br>1. Proof and applications of the identity $\sin^2A + \cos^2A=1$.<br>2. Only simple identities to be given. | • Uses Trigonometric identities to solve problems. | • Proves trigonometric identities using $\sin^2A + \cos^2A=1$ and other identities |
| **3.** | **HEIGHTS AND DISTANCES:**<br>**Angle of elevation, Angle of Depression.**<br>1. Simple problems on heights and distances. Problems should not involve more than two right triangles. Angles of elevation / depression should be only 30°, 45°, and 60°. | • Applies Trigonometric ratios in solving problems in daily life contexts like finding heights of different structures or distance from them. | • Find heights and distances in real life word problems using trigonometric ratios |
| | **UNIT VI: MENSURATION** | | |
| **1.** | **AREAS RELATED TO CIRCLES**<br>1. Area of sectors and segments of a circle.<br>2. Problems based on areas and perimeter /circumference of the above said plane figures. (In calculating area of segment of a circle, problems should be restricted to central angle of 60°, 90° and 120° only. | • Derives and uses formulae to calculate areas of plane figures. | • Visualises and evaluates areas of sector and segment of a circle |

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| **2.** | **SURFACE AREAS AND VOLUMES**<br>1. Surface areas and volumes of combinations of any two of the following: cubes, cuboids, spheres, hemispheres and right circular cylinders/cones. | • Visualises and uses mathematical thinking to discover formulae to calculate surface areas and volumes of solid objects (cubes, cuboids, spheres, hemispheres, right circular cylinders/cones, and their combinations). | • Evaluates the surface areas and volumes of combinations of solids by visualisation |
| | **UNIT VII: STATISTICS AND PROBABILITY** | | |
| **1.** | **STATISTICS**<br>1. Mean, median and mode of grouped data (bimodal situation to be avoided). | • Calculates mean, median and mode for different sets of data related with real life contexts. | • Computes the mean, of a grouped frequency distribution using direct, assumed mean and step deviation method.<br>• Computes the median and mode of grouped frequency distribution by algebraic method |
| **2.** | **PROBABILITY**<br>1. Classical definition of probability.<br>2. Simple problems on finding the probability of an event. | • Applies concepts from probability to solve problems on the likelihood of everyday events. | • Determines the probabilities in simple real-life problems |

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## MATHEMATICS-Standard (Code-041)

### QUESTION PAPER DESIGN

### CLASS – X (2025-26)

**Time : 3 Hours** **Max. Marks : 80**

| S. No. | Typology of Questions | Total Marks | % Weightage (approx.) |
|---|---|---|---|
| 1. | **Remembering :** Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.<br>**Understanding :** Demonstrate understanding of facts and ideas by organising, comparing, translating, interpreting, giving descriptions, and stating main ideas. | 43 | 54 |
| 2. | **Applying:** Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way. | 19 | 24 |
| 3. | **Analysing :** Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalisations.<br>**Evaluating :** Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.<br>**Creating :** Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions. | 18 | 22 |
| | **Total** | **80** | **100** |

| INTERNAL ASSESSMENT | **20 Marks** |
|---|---|
| • Pen Paper Test and Multiple Assessment (5+5) | 10 Marks |
| • Portfolio | 05 Marks |
| • Lab Practical (Lab activities to be done from the prescribed books) | 05 Marks |

**PRESCRIBED BOOKS:**

1. Mathematics - Textbook for class X - NCERT Publication
2. Guidelines for Mathematics Laboratory in Schools, Class X - CBSE Publication
3. Mathematics Exemplar Problems for Class X, NCERT Publication.

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